

TECHNICAL REPORT

Regarding Grazing of Cattle Guard of Train No. 64582 (DLI–HRF MEMU) with Unloaded Rail Panel at Km 1298/15 in HRS–HRF Section

1. Background

20 rail panels were received at HRS through DMT for unloading in the HRS–HRF section in connection with the proposed PQRS work. SSE/HRS, Shri Jayesh Sharma, was deputed for unloading of the panels along with a total of 27 staff.

On 17.09.2026, 12 rail panels were unloaded between Km 1298/18 and Km 1301/0. The remaining 10 panels were planned for unloading on 18.09.2026. A traffic block was allotted from 11:25 hrs to 15:25 hrs for the work.

During the unloading operation on 18.09.2026, the panels were unloaded from Km 1298/18 towards Km 1297/7. The traffic block was ultimately cleared at 16:50 hrs at HRS.

2. Incident

Train No. 64582 DLI–HRF MEMU was the first train to pass through the section after the unloading operation. The train departed HRS at 00:40 hrs. The Loco Pilot noticed an abnormal sound at approximately Km 1298/15.

On subsequent examination, an unloaded rail panel lying close to the running line at Km 1298/15 was found to have been grazed/touched by the cattle guard of the MEMU.

The train reached HRF at 01:09 hrs and departed HRF for HRS at 01:26 hrs. The train was stopped at Km 1298/16, the offending rail panel lying at Km 1298/15 was removed/cleared, and the train subsequently departed at 02:30 hrs and reached HRS at 02:43 hrs.

The full section was thereafter inspected and found safe for train movement.

3. Technical Cause of Grazing of Cattle Guard

The occurrence appears to have resulted from a **combined effect of the vertical level difference and the lateral proximity of the unloaded rail panel to the running line.**

Ballast had been unloaded in the HRS–HRF section on 10.09.2026. The concerned rail panel was unloaded on the shoulder over this recently unloaded ballast. Consequently, the unloaded rail panel was supported at a higher level than the surrounding/normal shoulder level.

Further, there was an additional difference in rail section. The existing running track at the location was laid with **52 kg rail**, whereas the newly unloaded rail panel was of **60 kg rail**. Therefore, the difference in rail section, together with the panel being supported over the recently unloaded ballast, resulted in a **net increase in the top level of the unloaded rail panel with respect to the existing running rail level.**

During passage of Train No. 64582, the cattle guard consequently came into contact/grazed the unloaded rail panel, producing the sound noticed by the Loco Pilot.

4. Contributory Factors

The following factors contributed to the occurrence:

1. **Recently unloaded ballast was present on the shoulder**, on which the rail panel was supported. This resulted in the panel being positioned at a higher level than normal shoulder-laid rail material.
2. **Difference in rail section:** the existing track was of **52 kg rail**, whereas the unloaded panel was of **60 kg rail**. The difference in rail section itself contributed to the relative vertical level difference.
3. The **combined vertical level difference due to shoulder ballast and difference in rail section** increased the effective height of the unloaded panel with respect to the running rail.
4. The panel was located sufficiently close to the running line that the increased height reduced the effective clearance available for the cattle guard.
5. As per the statement of SSE/HRS, the section was inspected after completion of unloading; however, the panel at Km 1298/15, being close to the running line but at level difference than running line, was not noticed during this inspection.

5. Safety Aspect

The incident highlights that while unloading rail panels, **both horizontal clearance from the running line and vertical level with respect to the running rail are required to be ensured.**

A rail panel unloaded on the shoulder should not be allowed to remain in such a position that:

- it encroaches into the dynamic clearance envelope of passing rolling stock;
- its end or any portion comes close to the running line;
- it is supported on newly unloaded ballast in a manner that substantially raises its level;
- the difference in rail section/height further increases the effective obstruction height.

Particular attention is required where the running track and unloaded panels are of different rail sections, such as 52 kg and 60 kg rails.

6. Corrective and Preventive Measures

The following measures are recommended:

1. The location for unloading should be inspected before unloading so that recently unloaded ballast or other material does not cause an unintended increase in the level of the panel.
2. In case of different rail sections, such as **52 kg and 60 kg rails**, the difference in rail section should be specifically considered while assessing the clearance.
3. After completion of unloading, the SSE/JE in-charge should conduct a **dedicated post-unloading inspection**, particularly from the perspective of the cattle guard/undergear clearance of the rolling stock.
4. The entire unloading stretch should be inspected from both directions to identify any panel lying unusually close to the running line.
5. Before permitting the first train after unloading, confirmation should be obtained from the supervisor-in-charge that all panels are safely positioned and clear of the moving-dimension envelope.

6. Staff engaged in unloading should be briefed that **safe unloading is not merely a matter of keeping the panel outside the track; its vertical level and proximity to the running line are also critical safety parameters.**

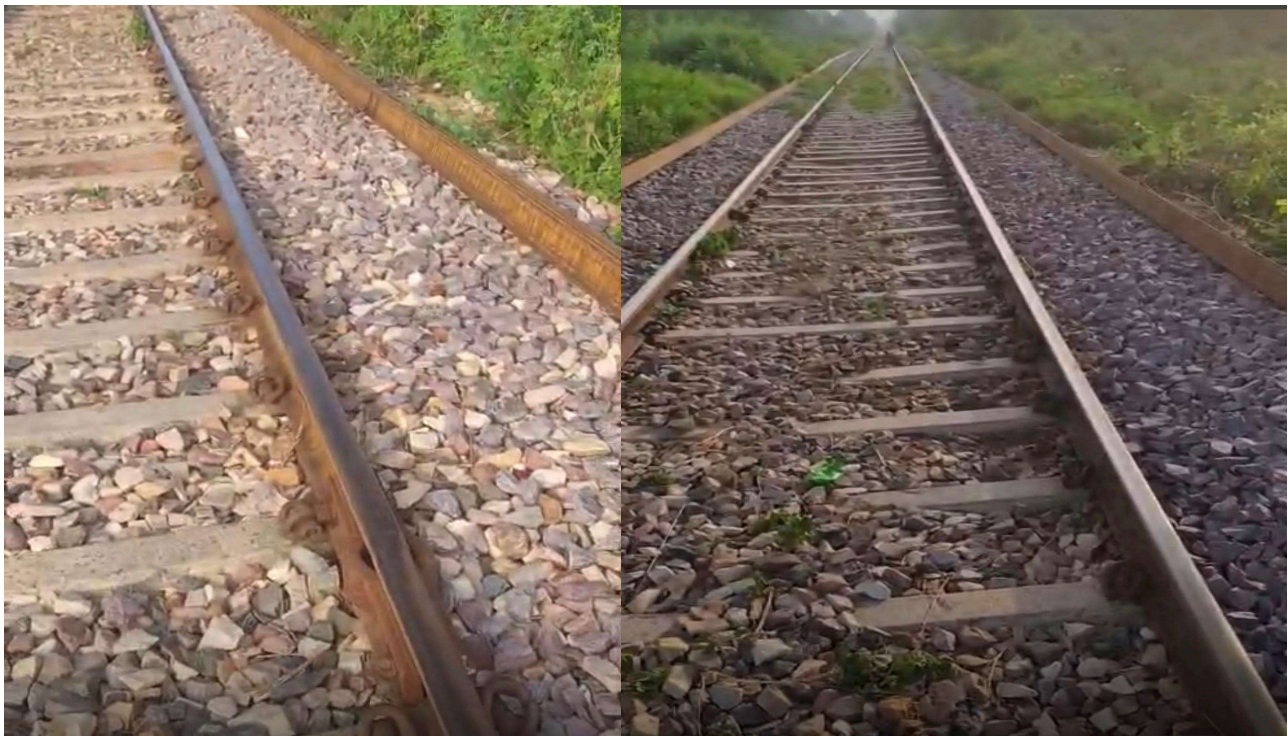
7. Conclusion

The grazing of the cattle guard of Train No. 64582 with the unloaded rail panel at Km 1298/15 appears to have occurred because the panel was lying close to the running line and at an elevated effective level.

The elevation was contributed by **(i) placement of the panel over recently unloaded shoulder ballast and (ii) the difference between the 60 kg unloaded rail section and the existing 52 kg running rail section.** The resulting net vertical level difference reduced the available clearance between the cattle guard of the passing MEMU and the unloaded panel.

The occurrence could have been prevented by ensuring adequate lateral clearance from the running line and by carrying out a thorough post-unloading inspection specifically considering the **horizontal and vertical clearance of all unloaded panels with respect to passing rolling stock.**

The full section was subsequently inspected and found safe. SSE/HRS is placed under suspension following the incident.



Unloaded rail panels



Cattle guard (No mark found)



vivo V30 Pro



23mm f/1.88 1/100s ISO796
09/19/2026, 06:41

Vertical damper with mark

Avilay
Yadav

Digitally signed by
Avilay Yadav
Date: 2026.09.19
15:01:33 +05'30'

SR.DEN IV PRYJ

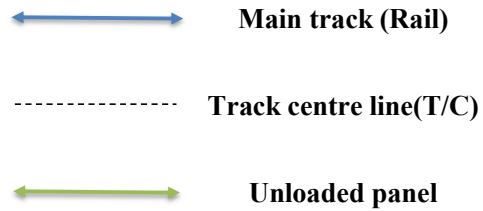
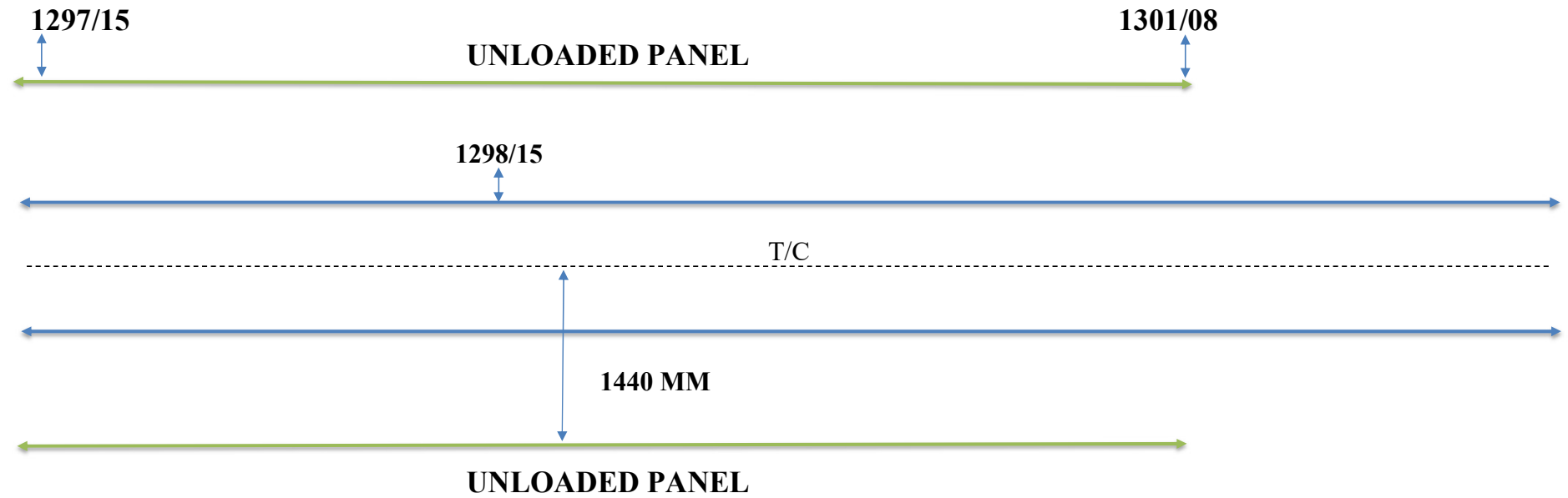
REHAN
SHAHID

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REHAN SHAHID
Date: 2026.09.19
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SR.ADEN/TDL

HRS (1297/2)

HRF (1305/11-13)



Schematic Diagram

MUKESH
KUMAR
SHARM
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Digitally signed
by MUKESH
KUMAR SHARMA
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17:55:48 +05'30'